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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December 2025

EE2231-1- ELECTRICAL ESTIMATION AND COSTING

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. An estimate provides:
A) Material cost
B) Labour cost
C) Total project cost
D) All of these
2. Which document gives rate and size of material to be used?
A) Specification
B) Tender
C) Estimate
D) Schedule
3. Which rule regulates supply and use of electrical energy?
A) IE Rules
B) ISI Standards
C) Tender Rules
D) IEC Rules
4. In wiring, distribution of energy in a building is done through:
A) Service mains
B) Sub-circuits
C) Main switch board
D) All of these
5. Casing and capping wiring is suitable for:
A) Wet locations
B) Fire-prone areas
C) Dry places
D) Underground wiring
6. The utilization factor is the ratio of:
A) Lumens on working plane to lumens emitted
B) Watts/lumen
C) Lumen/watt
D) Input power to output
7. The depreciation factor accounts for:
A) Losses in conductor
B) Ageing of lamp/loss of illumination
C) Labour charges
D) Cable resistance
8. The main switch in domestic wiring should be located:
A) Near the service entry
B) Away from the service entry
C) Next to each
D) On the roof
9. In conduit wiring, the main advantage is:
A) Cheapest
B) Mechanical protection & safety
C) Easy for temporary use
D) Less labour
10. The type of fuse used in house wiring:
A) HRC fuse
B) Kit-Kat fuse
C) Rewireable fuse
D) Cartridge fuse
11. Service line connecting supplier to consumer:
A) Distribution main
B) Service mains
C) Busbar
D) Sub-circuit
12. The load limit of a power sub-circuit is:
A) 1 kW
B) 2 kW
C) 3 kW
D) 10 kW
13. Starter suitable for squirrel cage induction motor up to 15 HP:
A) Auto-transformer
B) Star-delta
C) Direct On Line (DOL)
D) Rotor resistance

P.T.O.

14. The vertical distance between overhead line insulators in high roof buildings is:
 A) 35 cm B) 75 cm
 C) 100 cm D) 10 cm
15. The horizontal distance between two supports of overhead conductors is:
 A) Span length B) Pole length
 C) Ground clearance D) Wire length
16. The most commonly used substation in India:
 A) Indoor B) Outdoor
 C) Pole-mounted D) Foundation-mounted
17. The single line diagram represents:
 A) Detailed wiring B) 3-phase system symbolically
 C) Earthing system D) Insulator details
18. Earthing conductor in house wiring is generally made of:
 A) Aluminium B) Copper
 C) GI wire D) Both B & C
19. Tender is:
 A) Offer to do work at quoted rate B) Market survey sheet
 C) Comparative statement D) Payment bill
20. Comparative statement is used to:
 A) Compare tender quotations B) Compare wiring methods
 C) Compare conductor size D) Compare insulation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define estimation. Explain factors affecting estimation and importance.	8	L*1	1	1
b)	A hall 30 m × 15 m with 5 m ceiling height is to be illuminated to 120 lux. Coefficient of utilization = 0.5, depreciation factor = 1.3, efficiency of 40 lumen/W, 80 W lamp. Calculate number of lamps, their spacing mounting height, total wattage.	8	L3	2	2
2. a)	Write short note on Indian Electricity Act 1910 and IE Rules 1956.	8	L2	2	1
b)	Explain advantages, applications and types of conduit wiring.	8	L2	1	1
3. a)	What is tender? Explain the general guidelines for inviting tender.	8	L2	1	2
b)	List out the different types of wiring. Write advantages of multi stranded cables.	8	L2	2	1
Unit – II					
4. a)	Draw and explain underground service connection with parts.	10	L3	3	2
b)	Write a note on clearance of conductors from ground.	6	L2	4	1
5. a)	Explain important considerations regarding motor installation wiring?	6	L2	3	2
b)	Prepare estimate of materials and cost for giving 230 V, 5 kW single-phase service connection to a house, overhead line 25 m away.	10	L3	4	1
6. a)	Define the terms Cross arms, Span Lengths, Lightning Arrestors, Phase Plates, Danger Plates, Anti Climbing Devices, Bird Guards, Beads of Jumpers.	12	L2	3	2
b)	Write a note on earthing of transmission line.	4	L1	4	1
Unit – III					
7. a)	Explain substation earthing.	8	L2	5	1
b)	Draw Single Line Diagram of 11kV/400V of a 100 kVA indoor substation and list the materials used.	8	L3	5	1
8. a)	List main equipment of substation and their functions.	8	L2	5	1
b)	Classify substations. Compare indoor and outdoor substations.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
